

team at University of Arkansas. It lets you detect brain injury using a network of flexible sensors. Collected data is then sent through ZigBee and Bluetooth to a receiver.

In 2014, Novartis also announced that its eye-care division, Alcon, would soon license its smartlens technology for all ocular medical uses. This technology involves the use of non-invasive sensors, microchips and other miniaturised electronics that are embedded into the contact lens. A LinkedIn search on Alcon showed the profile of an electronics and software lead engineer at Alcon, which had Institute of Electrical and Electronics Engineers (IEEE) 802.15.6 standard listed. What is that?

LAN, WAN and, now, BAN

After local area network (LAN) and wide area network (WAN), you now have body area network (BAN). Also known as body sensor network, this is effectively a wireless network of wearable computing devices and implantables.

IEEE 802.15 Task Group 6 (BAN) is developing a communication standard optimised for low-power devices and operation on, in or around the human body (but not limited to humans) to serve a variety of applications including medical, consumer electronics, personal entertainment and others.

New memory for wearables

Targeting such wearable devices for applications like hearing aids, pulse meters and activity trackers, Fujitsu has introduced their new 1Mbit serial ferro-electric random access memories (FRAMs). The new FRAM memory delivers 77 per cent reduction in surface mount area, apart from contributing to longer battery life by minimising power consumption during write operations. The FRAM developed by Cypress is immune to corruption by magnetic fields and radiation, thus allowing for its use in medical wearables for capturing data

instantly with complete security.

Maxim's newly-designed gamma resistant non-volatile memory based on 1-wire technology allows calibration of consumable medical sensors, tools and accessories to their host medical instrument in the field.

Meeting standards

Got an amazing idea for a wearable device that you believe will change the world? Figured out all the hardware and software design, too? If you want to see your device enter the market and start selling, you have to ensure that it meets medical standards. Do not despair though, because Shreekant Pawar, co-founder and CMO at Diabeto, has just the advice you need.

"When we were developing Diabeto, in the early stages, we were completely unaware of medical standards; we did not even know what standards were. Only when we developed the first prototype, one investor asked us if our design was according to the standards. Just like us, many medical devices start-ups are completely unaware about the prevailing medical standards," says Pawar.

He explains that, broadly there are two types of standards, namely, vertical and horizontal. Typically, a vertical standard is specific to a product or a device, while a horizontal standard applies to a wide range of devices. International Organization for Standardization (ISO) 13485 and ISO 14971 are examples of the most important horizontal standards that are widely used.

ISO 13485 is optional, but is considered the *de facto* standard for companies that sell in Europe. One good thing about ISO 13485:2003 is that, it is made specifically for medical devices and is accepted worldwide, except the USA. The US FDA follows a different system. Having said that, there is 90 per cent overlap between ISO 13485 and the US FDA system.

ISO 14971 covers risk management. Your device needs to pass

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